Dataset Cards

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Responsible Plant Classifier

THE CONNECTED WEEK

One current dataset-card record.

Write supported entries, repair guesses and update counts without inventing source identities.

Keep one habit: Fresh W1–W3/W5 fields → selected independent card → versioned project artifact → delayed return.

MAIN LESSON · SECTION 1

Recall · Retrieve before notes

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Leakage. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Label Quality. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Classification or Regression?. Give an example and a mistake that the idea helps you avoid.

Leakage

> **My remembered idea, example and correction**

Label Quality

> **My remembered idea, example and correction**

Classification or Regression?

> **My remembered idea, example and correction**

W1 · Choose evidence-linked fields

> Current dataset-card questions use fresh response fields; earlier audit responses remain review.

Complete current card facts

Photo practice v1: teacher-created fictional classroom fixture;18photos,P1–P6 witha/b/c views each;6plants,one tablet,indoor fluorescent. Label vocabulary visible spot/no visible spot/review; per-photo assignments/history not supplied. Exact date, external-reuse terms, split assignments and quality checks not supplied. No model evaluation.

v2 addsP1d–P6d outdoor photos:24photos/6plants,18indoor/6outdoor; added-view device/lighting details unknown. Preserve earlier version and record the transformation.

Current task and response

- Name purpose/source/version, row/group counts, conditions, labels, split status and excluded use for v1. Mark at least two unknown fields.

> **My dataset-card reasoning and evidence**

▼ **Review · Earlier audit task and saved response**

For Robot practice v1, state what one row is intended to represent, the source, the unit and the meaning of None.

My reasoning and evidence

MAIN LESSON · SECTION 13

W2 · Write and update the card

> Current dataset-card questions use fresh response fields; earlier audit responses remain review.

Complete current card facts

Photo practice v1: teacher-created fictional classroom fixture;18photos,P1–P6 witha/b/c views each;6plants,one tablet,indoor fluorescent. Label vocabulary visible spot/no visible spot/review; per-photo assignments/history not supplied. Exact date, external-reuse terms, split assignments and quality checks not supplied. No model evaluation.

v2 addsP1d–P6d outdoor photos:24photos/6plants,18indoor/6outdoor; added-view device/lighting details unknown. Preserve earlier version and record the transformation.

Current task and response

- From Photo practice v1's complete manifest, fill purpose/source/version, photo and plant counts, known conditions, label status and split status. Mark one field not supplied and one use with no supporting evaluation. Then update the card for v2's P1d–P6d outdoor additions, preserving the earlier version.

My dataset-card reasoning and evidence

▼ **Review · Earlier audit task and saved response**

Explain each number in 5 4 2 1. How many extra ID occurrences are there, and how many rows belong to a repeated-ID group?

My reasoning and evidence

MAIN LESSON · SECTION 14

W3 · Repair guessed entries

> Current dataset-card questions use fresh response fields; earlier audit responses remain review.

Complete current card facts

Photo practice v1: teacher-created fictional classroom fixture;18photos,P1–P6 witha/b/c views each;6plants,one tablet,indoor fluorescent. Label vocabulary visible spot/no visible spot/review; per-photo assignments/history not supplied. Exact date, external-reuse terms, split assignments and quality checks not supplied. No model evaluation.

v2 addsP1d–P6d outdoor photos:24photos/6plants,18indoor/6outdoor; added-view device/lighting details unknown. Preserve earlier version and record the transformation.

Current task and response

- A dataset card has every heading filled, but each unknown field was guessed. Does completeness establish that its claims are correct or that a model is ready? Name the repair.

My dataset-card reasoning and evidence

▼ **Review · Earlier audit task and saved response**

A learner changes None to 0 and -2 to 2, then reports a clean dataset. Explain why this is not a justified repair and what record should be kept.

My reasoning and evidence

MAIN LESSON · SECTION 16

W5 · Check a proposed reuse

> Current dataset-card questions use fresh response fields; earlier audit responses remain review.

Complete current card facts

Photo practice v1: teacher-created fictional classroom fixture;18photos,P1–P6 witha/b/c views each;6plants,one tablet,indoor fluorescent. Label vocabulary visible spot/no visible spot/review; per-photo assignments/history not supplied. Exact date, external-reuse terms, split assignments and quality checks not supplied. No model evaluation.

v2 addsP1d–P6d outdoor photos:24photos/6plants,18indoor/6outdoor; added-view device/lighting details unknown. Preserve earlier version and record the transformation.

Current task and response

- A reader wants to use v1 for outdoor growth prediction. Name which supplied facts are relevant, which labels/conditions/permission questions are unanswered and why neither photo count nor a completed card establishes suitability.

My dataset-card reasoning and evidence

▼ **Review · Earlier audit task and saved response**

Someone duplicates each row ten times and claims the export now covers more kinds of trips. Explain what changed and what did not.

My reasoning and evidence

Try alone · Selected experiment task

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Only the listed20photos/5plants/4views each,one phone,one indoor setting and absent growth labels are supplied. Do not invent the creator, date, permission, species, class balance or growth outcomes. Record unknown source fields as questions, propose a label contract and plant grouping, and distinguish proposals from completed assignments or evaluation.

  A claim about growth needs a defined target and observations, not just photos.

- Write key dataset-card entries for20 photos: five plants, four views each, one phone, one indoor setting. Growth labels have not been supplied.

- Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Independent case: Changed independent case

> **My independent explanation, trace and limits**
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MAIN LESSON · SECTION 9

Project · Keep the experiment evidence

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Publish the Plant Classifier **dataset card** before model experiments begin; update it when data changes.

- The project change and **evidence** I will keep

> **Write or draw your reasoning and evidence.**

MAIN LESSON · SECTION 17

Feedback · Repair and return

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	

Evidence	My record
One remaining question	

MAIN LESSON · SECTION 10

Finish · Judge your evidence

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- - I can explain the mechanism.
  - I can show evidence.
  - I can handle a changed case.
  - I can name a limit.

  Mark each as not yet, with help or independently, and attach supporting work.

- Teacher review _____ Next step _____

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Earlier trip-audit completion record: retain its saved response as review. Use current W2 for this week's dataset-card writing and version update.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

▼ Extra practice · Individual reasoning check

REFERENCE · SECTION 3

Extra practice · Individual reasoning check

> Keep this task’s complete inputs, stated rule and support record together.

Task and evidence record

- Use the worked case for Dataset inspection and cards. Proposed explanation: “A complete card certifies data quality”. Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Earlier learning target

REFERENCE · SECTION 5

Reference · Earlier learning target

> Keep this task’s complete inputs, stated rule and support record together.

Task and evidence record

- Write five dataset-card fields for a leaf dataset and explain why '1000 images' alone is insufficient.

▼ **Extra practice · Earlier complete case**

REFERENCE · SECTION 6

Extra practice · Earlier complete case

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- A **dataset card** lists fields but omits source and intended use.
- Add information needed to assess suitability.
- Show your trace, calculation or **evidence**

Write or draw your reasoning and evidence.

▼ Extra practice · Changed case

REFERENCE · SECTION 7

Extra practice · Changed case

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- A **dataset card** lists 500 images but omits collection conditions and intended use. Add two essential entries and explain their value.

- My independent answer and reason

> **Write or draw your reasoning and evidence.**

Week 9 Test and reflect

▼ Extra practice · Design a revealing test

REFERENCE · SECTION 8

Extra practice · Design a revealing test

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

▼ **Extra practice · Earlier transfer**

REFERENCE · SECTION 11

Extra practice · Earlier transfer

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Write five dataset-card fields for a leaf dataset and explain why '1000 images' alone is insufficient.

- Record support used: none, hint or teacher explanation.

My independently reasoned answer

▼ **Reference · Earlier W4 record**

REFERENCE · SECTION 15

Reference · Earlier W4 record

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Schema for this exercise: one intended row per distinct trip ID; each row is an (ID, duration) pair. Duration is numeric minutes or None for unmeasured. Negative durations are invalid; a recorded zero is allowed. Count extra ID occurrences beyond the first and do not silently alter the raw data.

- A new fictional indoor trip export has (P,4), (Q,None), (Q,None), (R,-1), (S,6), (T,0), with durations in minutes and the same **schema**.

  Count raw rows, distinct IDs, missing durations, negative durations and extra ID occurrences. Identify overlap. Draft a short **dataset card** stating source, row meaning, unit, findings, unchanged version and one excluded use.

My reasoning and evidence

▼ **Extra practice · Pictured case**

REFERENCE · SECTION 18

Extra practice · Pictured case

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Build a **dataset card** that reveals what the photo count hides.
- Teacher-created dataset: 18 photos of 6 fictional plants, 3 views each, one tablet, indoor fluorescent light. Label rule is visible spot/no visible spot/review.
- All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.
- What two counts should the revised card show? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.
- Support used: alone / hint / worked example. This record is practice **evidence**.

- Project connection: For your Responsible Plant Classifier, save the **input**, procedure, observed result and **limitation** of this check.

  Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.



Follow the labelled picture with the matching explanation. Open the vector diagram

My picture-based explanation and changed-case prediction

▼ **Reference · Project portfolio milestone**

REFERENCE · SECTION 19

Reference · Project portfolio milestone

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Project: Responsible Plant Classifier
- Current milestone: Make a transparent first version. Keep a trace of how one **input** becomes an **output**; explain the rule or learned component.
- For your Responsible Plant Classifier, save the **input**, procedure, observed result and **limitation** of this check.

  Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	

Evidence to keep	My record
One change since the previous checkpoint and why	
One error or limit and the next test	

> **Explain what this week adds to your project. Reuse last week's record rather than starting again.**

▼ Extra practice · Predict, check and repair

REFERENCE · SECTION 20

Extra practice · Predict, check and repair

> Keep supported feedback separate from the reserved independent case.

First supported question and hint

- Build a **dataset card** that reveals what the photo count hides.
- Which labelled step matches this detail? Split by plant; state that outdoor use and disease diagnosis have not been evaluated.
- 1. Record limits
- 2. Describe units

- 3. Describe source

- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ **Check the first explanation**

After your attempt, compare with the supported explanation: Record limits Split by plant; state that outdoor use and disease diagnosis have not been evaluated. Read the picture in order.

Split by plant; state that outdoor use and disease diagnosis have not been evaluated. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Changed-case question and hint

- Changed-case practice: The dataset adds 6 outdoor photos of the same plants. Which conclusion is supported by the supplied facts?

- 1. We can decide the result without examining the changed **input** or the procedure.

- 2. Photo count becomes 24, plant count remains 6; lighting coverage changes.

- 3. This one example guarantees correct results for every future **input**.

- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ **Check the changed explanation**

After your attempt, compare with the supported explanation: Photo count becomes 24, plant count remains 6; lighting coverage changes. Additional views broaden conditions without creating new independent plant identities.

Repair and keep the record

Review Leakage (ai-4/8). Goal: Choose a split unit that matches the intended **prediction** claim. Short practice: How many independently new plant identities are in the original test?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Label Quality (ai-4/6). Goal: Write an observable label rule with boundaries and a review option. Short practice: Why should two annotators not vote away Z's missing **evidence**?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Classification or Regression? (ai-4/3). Goal: Identify a task from the meaning of its target. Short practice: Did regression MAE improve when the boundary changed?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

> **What I repaired, the evidence I kept and the next test**
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▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Predict, run and debug

Keep your completed work

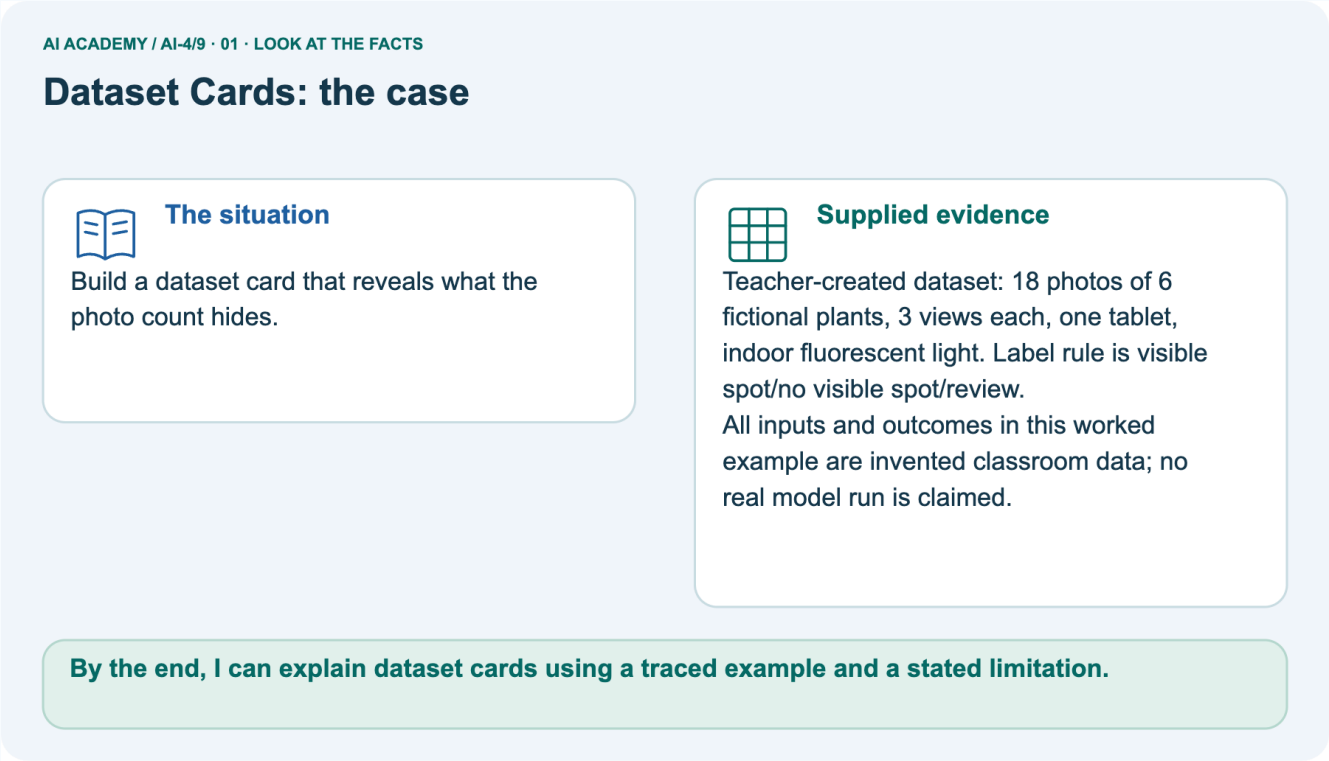
▼ **Reference · Six-picture response sheets**

REFERENCE · SECTION 21

Reference · Six-picture response sheets

> Choose the relevant picture instead of repeating every task.

01. Dataset Cards: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

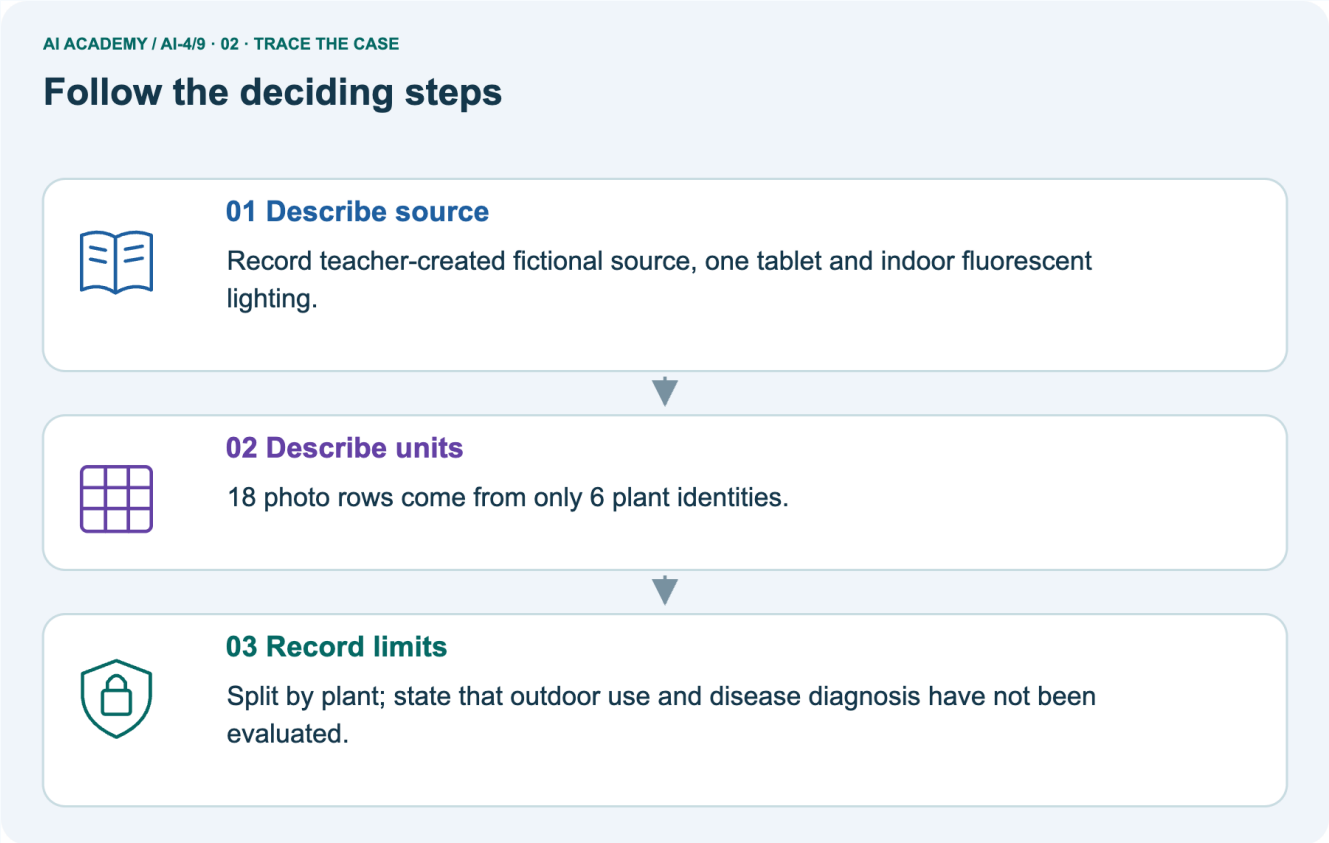
Build a **dataset card** that reveals what the photo count hides. Supplied facts: Teacher-created dataset: 18 photos of 6 fictional plants, 3 views each, one tablet, indoor fluorescent light.

Label rule is visible spot/no visible spot/review. All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Describe source: Record teacher-created fictional source, one tablet and indoor fluorescent lighting. → Describe units: 18 photo rows come from only 6 plant identities. → Record limits: Split by plant; state that outdoor use and disease diagnosis have not been evaluated.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Turn an observation into a documented action

AI ACADEMY / AI-4/9 · 03 · INSPECT THE LESSON MODEL

Turn an observation into a documented action

Finding	What it establishes	Next action
B appears twice	Identity needs review	Check export history
Two None values	2 of 5 rows lack duration	Keep unknown explicit
C has minus 2	Invalid elapsed duration	Check source and units

Flags can overlap: both rows with ID B also have missing duration. Do not add flag counts as if they were separate groups.

Follow the labelled picture with the matching explanation. Open the vector diagram

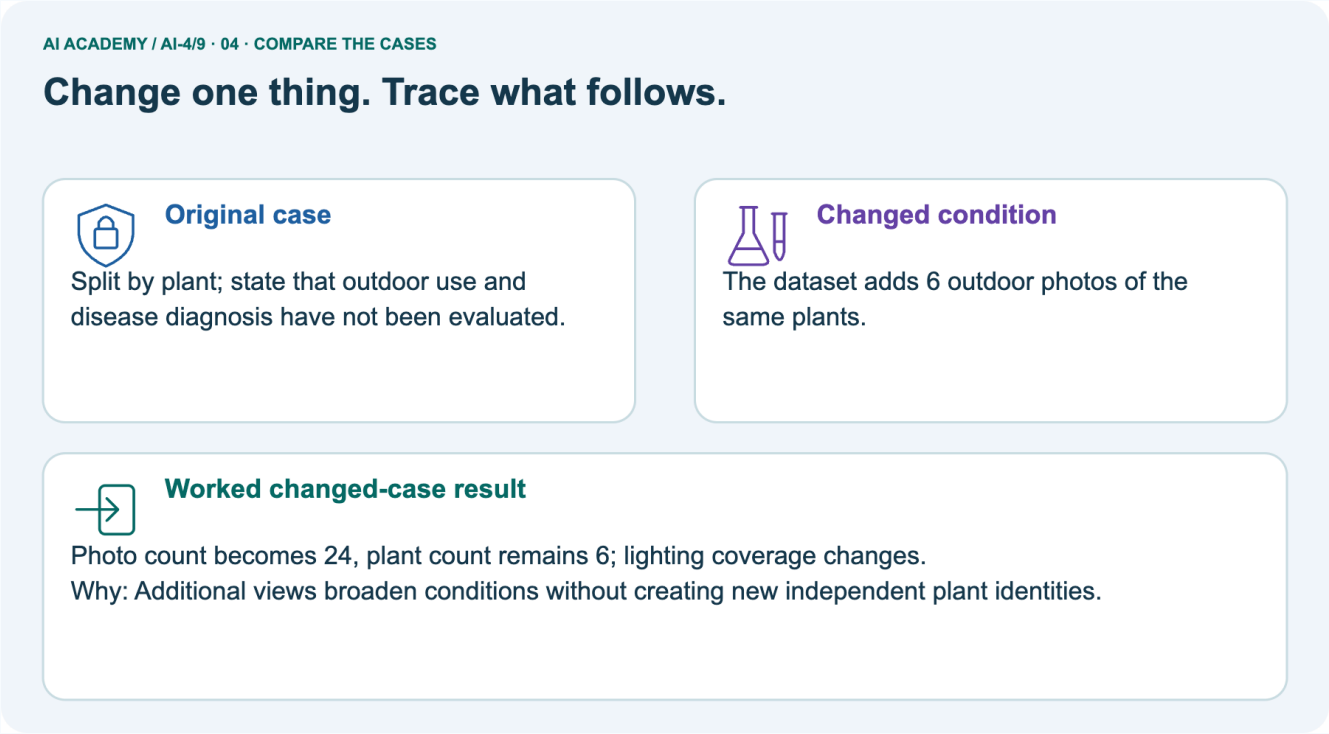
Turn an **observation** into a documented action. B appears twice | Identity needs review | Check export history. Two None values | 2 of 5 rows lack duration | Keep unknown explicit.

C has minus 2 | Invalid elapsed duration | Check source and units

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

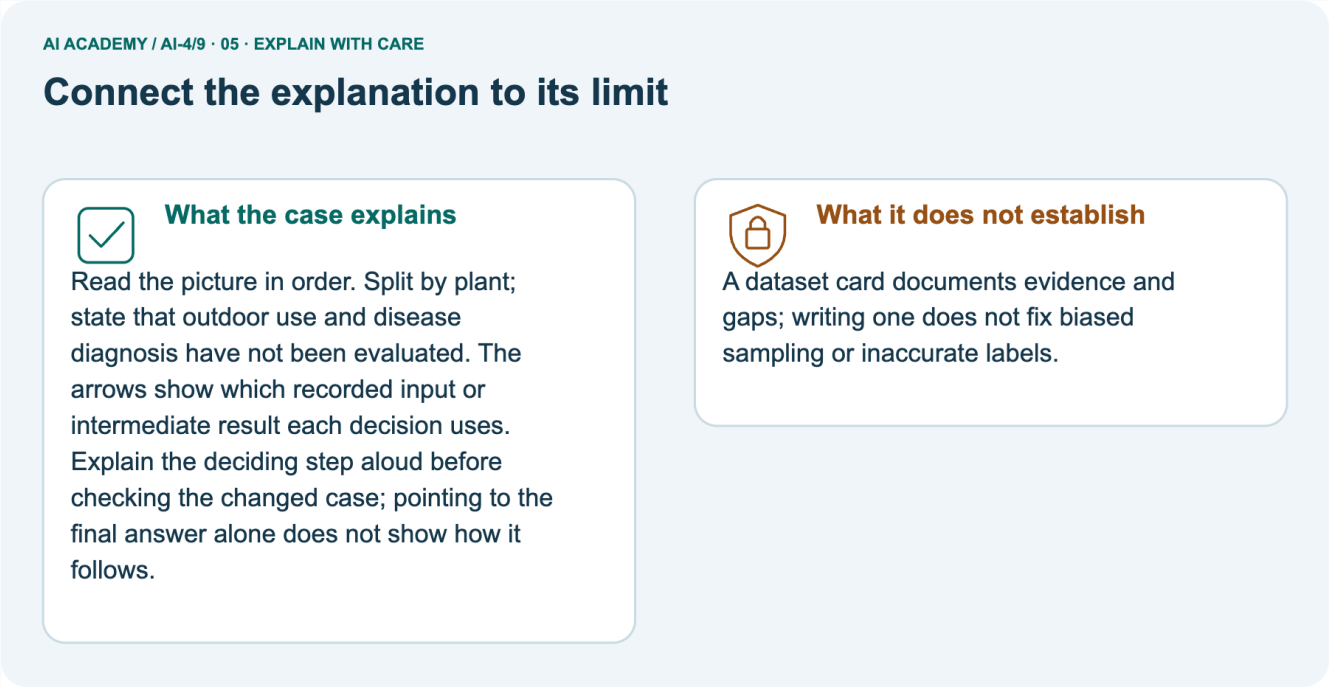
Original case: Split by plant; state that outdoor use and disease diagnosis have not been evaluated. Changed condition: The dataset adds 6 outdoor photos of the same plants.

Expected result: Photo count becomes 24, plant count remains 6; lighting coverage changes. Reason: Additional views broaden conditions without creating new independent plant identities.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture in order. Split by plant; state that outdoor use and disease diagnosis have not been evaluated. The arrows show which recorded **input** or intermediate result each decision uses.

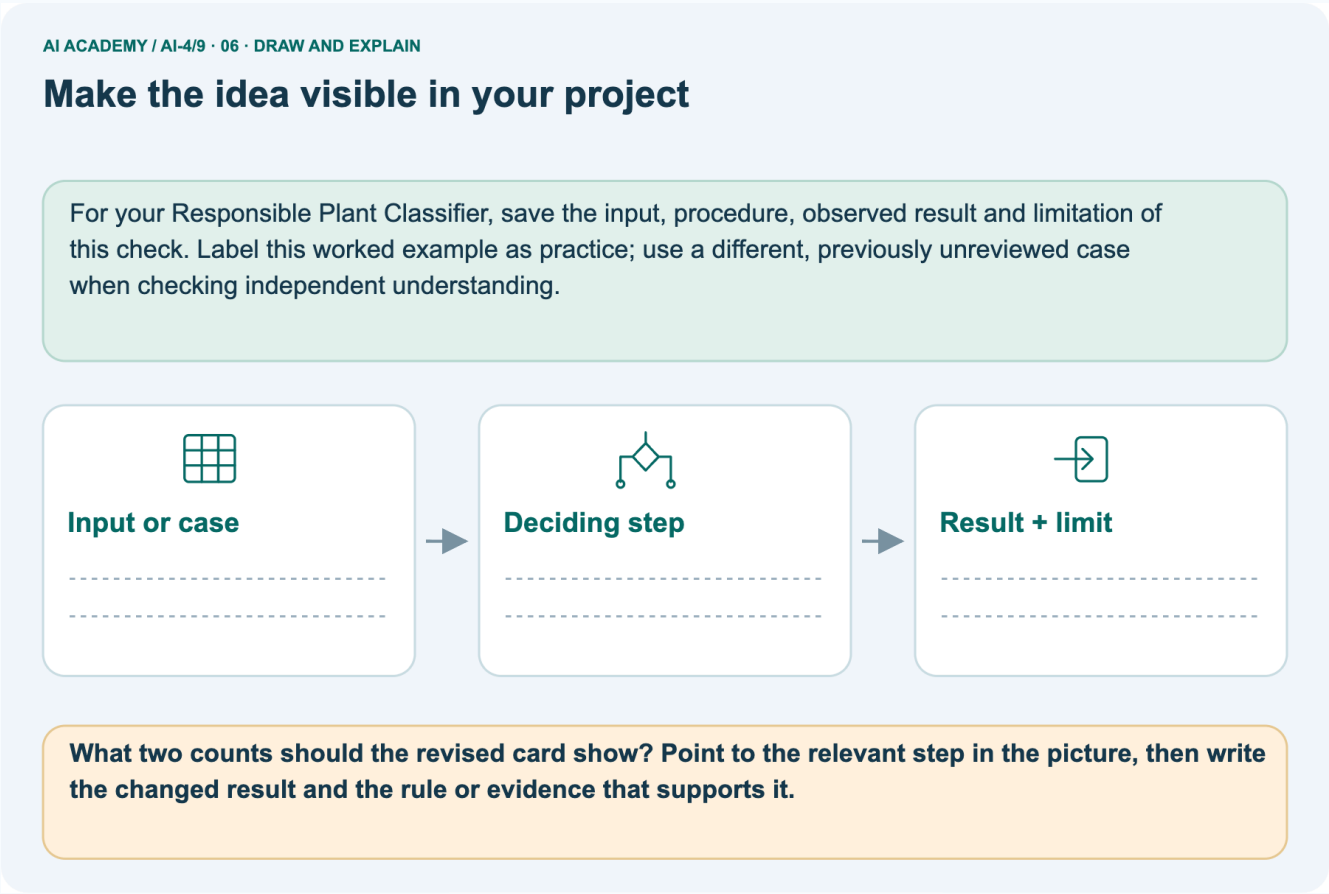
Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Limit: A **dataset card** documents **evidence** and gaps; writing one does not fix biased sampling or inaccurate labels.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Responsible Plant Classifier, save the **input**, procedure, observed result and **limitation** of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: What two counts should the revised card show? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

▼ Earlier wording, alternatives and picture transcripts

01. Dataset Cards: the case

02. Follow the deciding steps

03. Turn an observation into a documented action

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.